

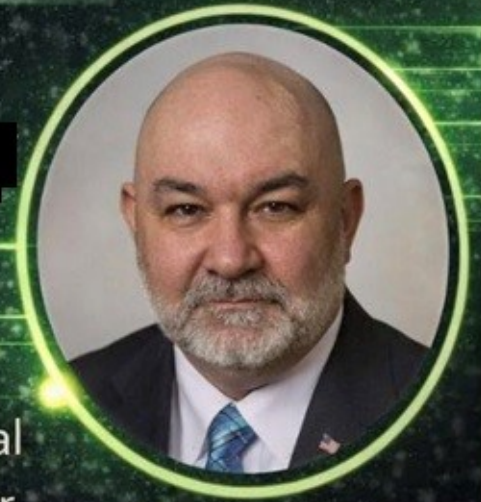
Harvest Now, Decrypt Later

Presentation

Chicago Quantum Group
July 2026

William Slater

Senior Cybersecurity Professional
& Quantum Technology Manager



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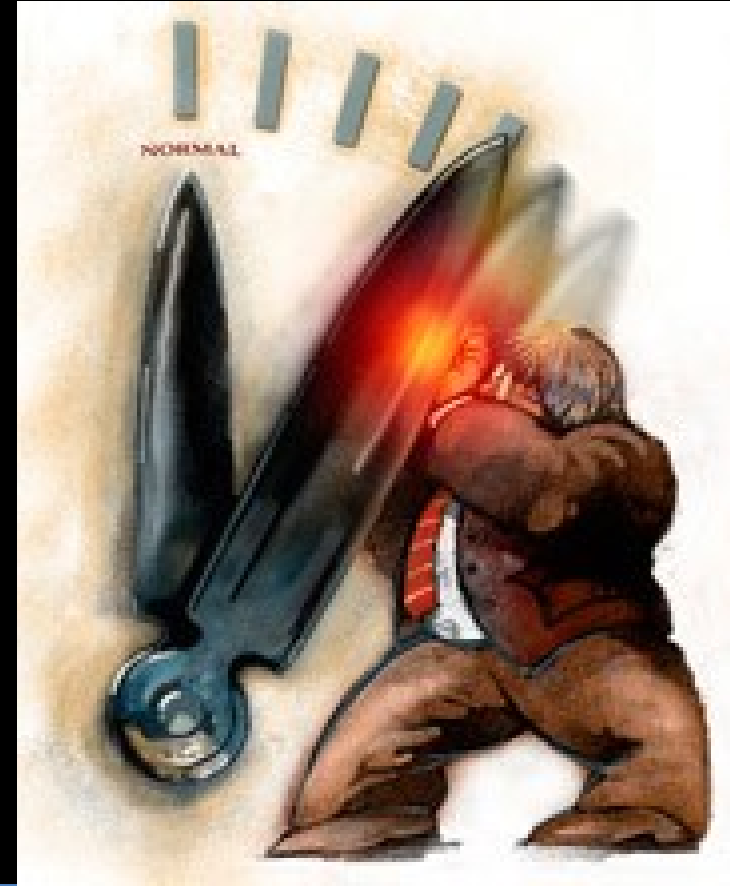
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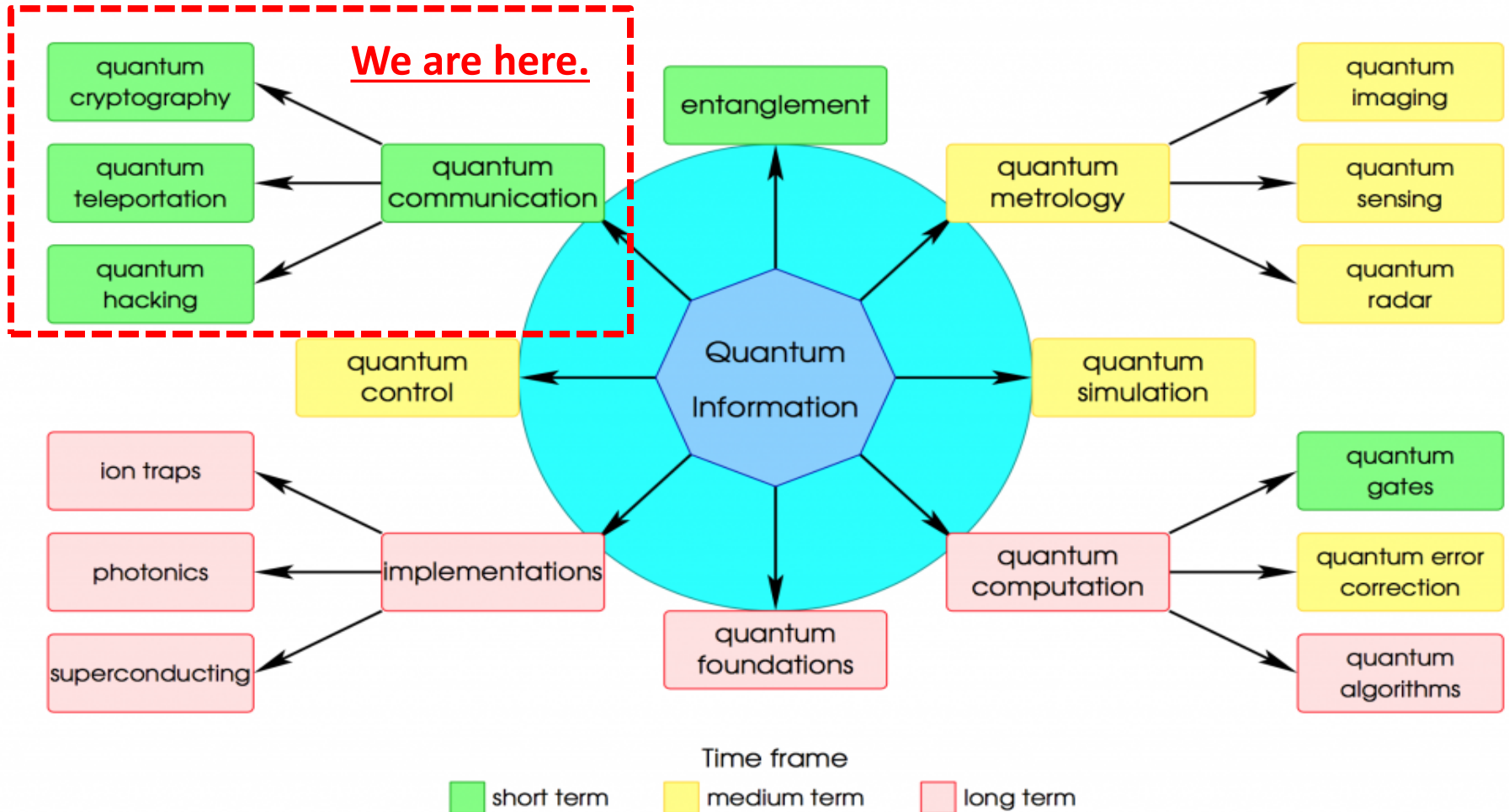
- Harvest Now, Decrypt Later– 2026 (July 2026)
- Presentation
 - <https://billslater.com/hndl2026.pdf>

Agenda

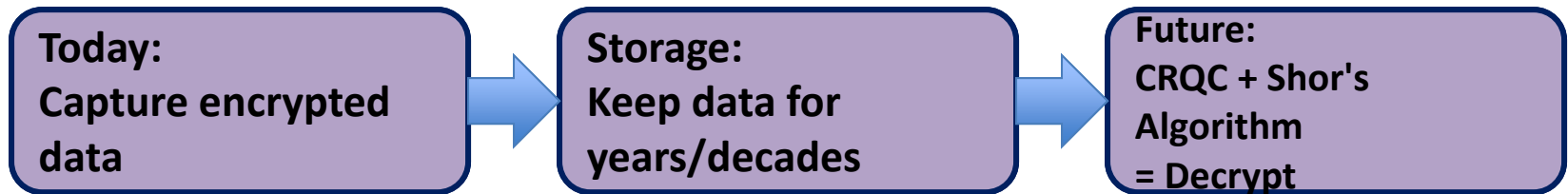
- What Is HNDL?
- Why HNDL Is Serious
- An Internet Minute
- Quantum Computers and Shor's Algorithm
- PQC as a Defense Against HNDL
- Illustrative PQC Readiness
- Who Has Started their PQC Migrations
- PQC Readiness with Timeline
- Key Mitigation Steps to Reduce HNDL Risk
- HNDL Mitigation Lifecycle
- Illustrative Quantum Risk Timeline
- Cryptographic Agility
- Key Takeaways and a Call to Action
- Questions and Answers
- References



Enter the Quantum Zone



HNDL in One Picture



HNDL = Adversaries harvest encrypted data now, store it and decrypt it later when quantum computers become powerful enough.

What is Harvest Now, Decrypt Later (HNDL)?

- Adversaries capture encrypted traffic and data today.
- They store it for years or decades, waiting for stronger compute.
- A future cryptanalytically relevant quantum computer (CRQC) can use Shor's Algorithm to break RSA/ECC keys.
- Result: Long-term confidentiality is lost even if systems appear secure today.

What Is Harvest Now, Decrypt Later? (HNDL)

- Data that traverses the Internet is usually encrypted.
- The Bad Guys and Nation States are capturing and storing this data that is encrypted using older encryption algorithms.
- Between 2028 and 2030, when CRQC computers become commonplace, the harvested data can be decrypted at the time and place of the Bad Guys' choosing and they use this decrypted data to their advantage.
- Note that the decrypted data can also be data that belongs to organizations that have completed their PQC Migrations, because the harvest data was encrypted using olded algorithms that are vulnerable to cryptoanalytic attacks of Quantum Computers.

What Does a Harvest Now, Decrypt (HNDL) Attack Look Like?

- When Q-Day occurs, data that has been encrypted and captured will be at risk. This risk is known as “Harvest Now Decrypt Later”.
- Do an Internet Search on “**Enron Email Dataset**” and also “**Peter Attia Epstein**” and you will understand the awkward situations involving unintended releases of data that was previously thought to be CONFIDENTIAL and PROTECTED.

Introduction to Enron Email Dataset

- Federal Energy Regulatory Commission (FERC) posted the Enron email dataset on the web in May of 2002
 - 619,446 emails
- Professor Leslie Kaelbling from MIT purchased the dataset
- SRI - integrity and security
- Professor William W. Cohen - CMU dataset
 - 150 user folders
 - 517,431 emails
 - 400Mb

6/8/2007

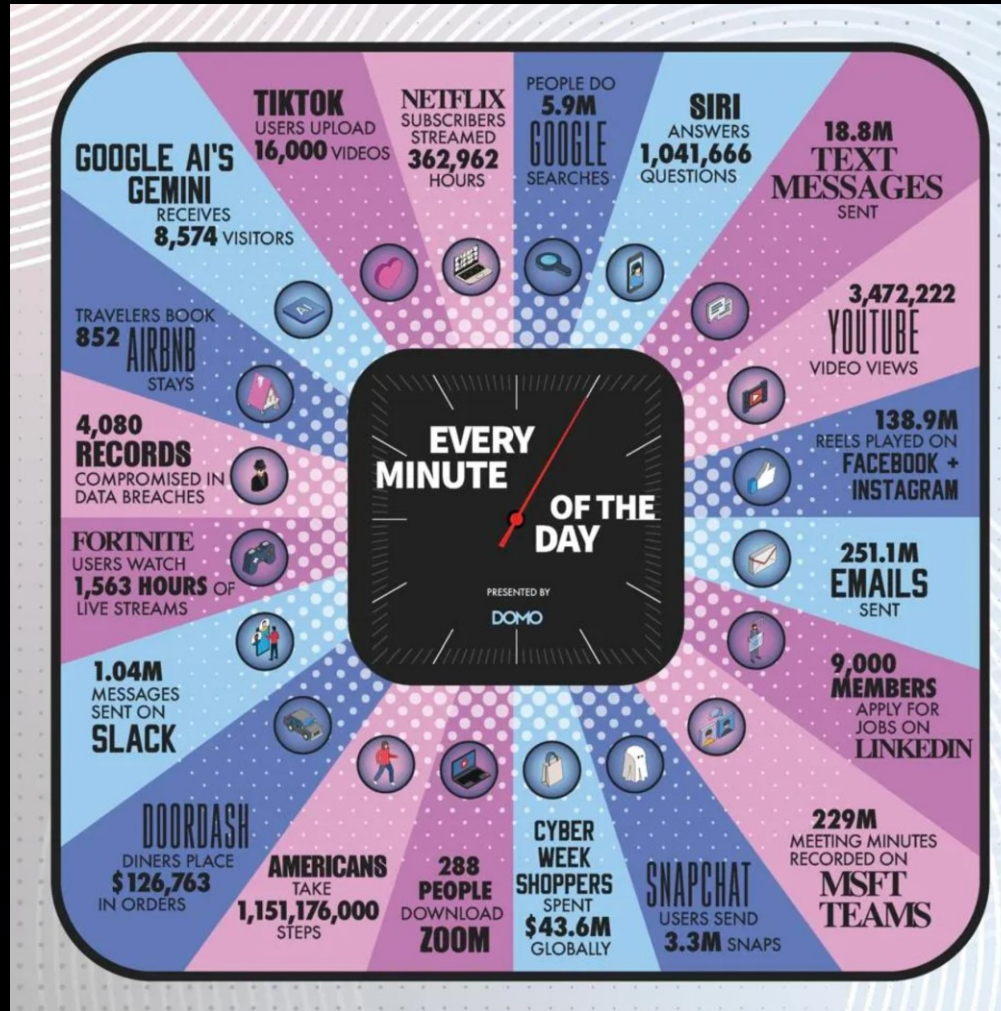
NAACSOS 2007

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The Eastern Herald
**Peter Attia Quits CBS as
Emails Spark Elite Scandal**
6 days ago

Internet Minute - 2025



Source:
<https://bondhighplus.com/blog/what-happens-in-an-internet-minute/>

Internet Minute – 2015 to 2025



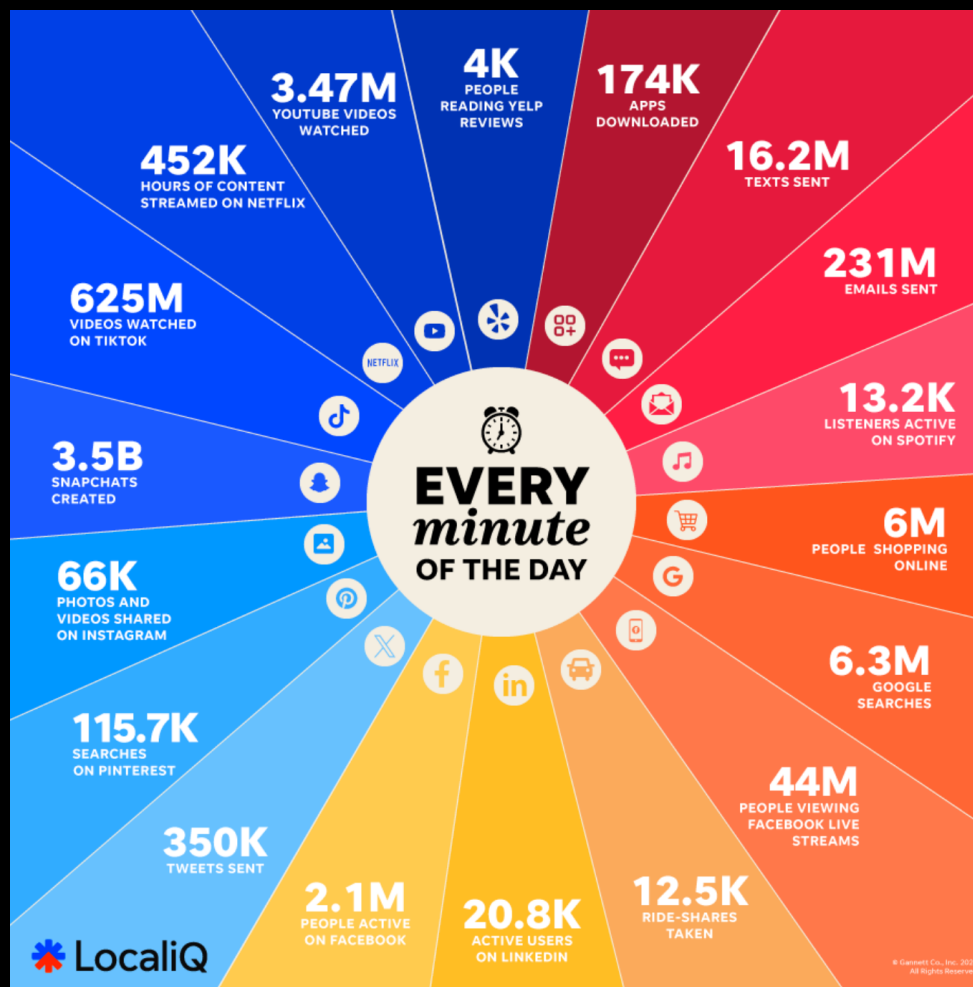
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Year	Emails Sent	Google Searches	YouTube Views	TikTok Views	Online Spend	App Downloads	Instagram Activity
2025	231M+	6.3M+	3.47M	625M videos watched	\$43.6M+	174K	66K photos shared
2024	231M	6.3M	3.47M	625M videos watched	\$43.6M (peak minute)	174K	66K photos shared
2023	231M	5.9M	3.67M	167M videos watched	\$1.5M+ (est.)	174K	694M songs streamed
2022	231M	5.9M	508 hrs uploaded	1B+ interactions	\$443K (Amazon only)	437K transfers	527K photos shared
2021	225M (est.)	5.7M (est.)	5.0M (est.)	167M (est.)	\$1.3M (est.)	450K (est.)	800K (est.) scrolling
2020	190M	4.1M	4.7M	N/A	\$1.1M	400K	694K scrolling
2019	188M	3.8M	4.5M	N/A	\$996K	390K	347K scrolling
2018	187M	3.7M	4.3M	N/A	\$862K	375K	174K scrolling
2017	156M	3.5M	4.1M	N/A	\$751K	342K	46K posts uploaded
2016	150M	2.4M	2.78M	N/A	\$203K	51K	527K photos shared
2015	204M	2M+	1.3M	N/A	\$83K	47K	3,000 uploads

Source:
<https://bondhighplus.com/blog/what-happens-in-an-internet-minute/>



Internet Minute - 2025



Source:
<https://localiq.com/blog/what-happens-in-an-internet-minute/>

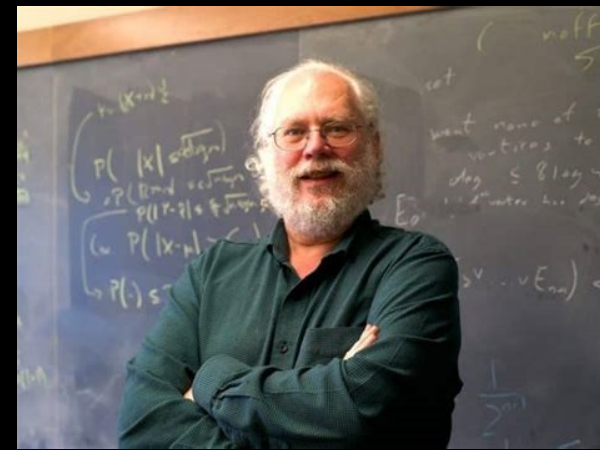
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Why HNDL is a Serious Privacy Threat

- Targets data with long sensitivity lifetimes: PII, medical records, biometrics, financial histories, IP, classified data.
- Once decrypted, historical communications can be searched, correlated, and profiled without the victim ever knowing.
- Bulk HNDL harvesting enables mass surveillance and retroactive exposure of past behavior.
- Trust in digital records, contracts, and identities can be undermined when historic data is silently decrypted.

Quantum Computers & Shor's Algorithm

- Shor's Algorithm can efficiently factor large integers and solve discrete log problems.
- This breaks widely deployed public-key schemes such as RSA, Diffie–Hellman, and ECC.
- A CRQC could decrypt stored sessions protected only by these algorithms.
- Symmetric algorithms (e.g., AES) are less impacted, but key sizes may need to increase.



Peter Shor
MIT Professor
Inventor of Shor's Algorithm

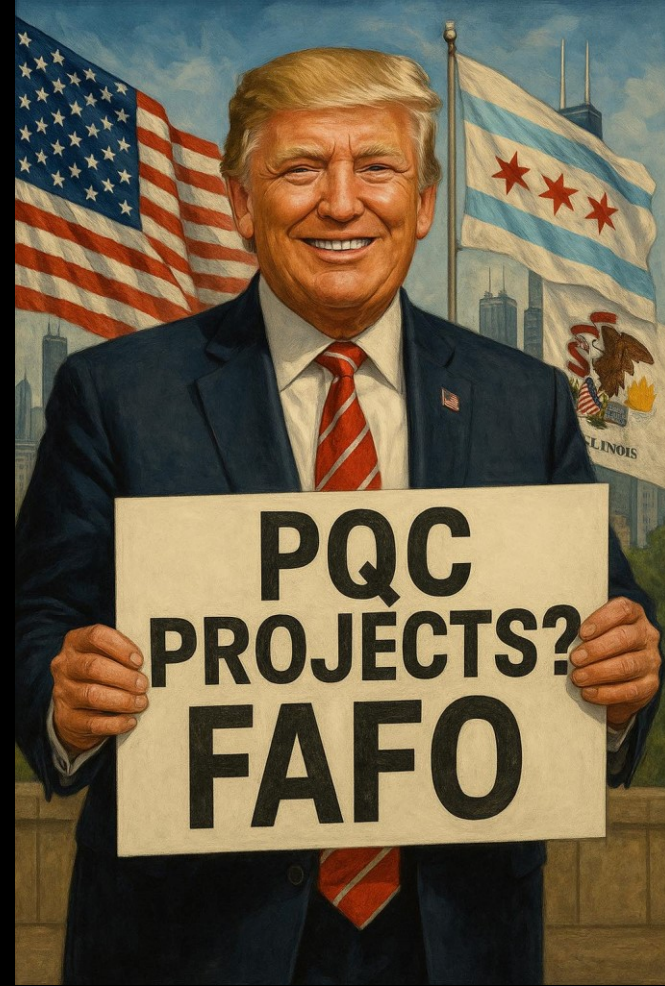
PQC as a Defense Against HNDL

Post-Quantum Cryptography (PQC) algorithms are designed to resist both classical and quantum attacks.

NIST has standardized PQC algorithms (e.g., Kyber for key establishment, Dilithium and SPHINCS+ for signatures).

NSA CNSA 2.0 and other national standards require quantum-resistant algorithms by the early 2030s.

Encrypting with PQC today ensures that harvested ciphertext remains secure even after Q-Day.

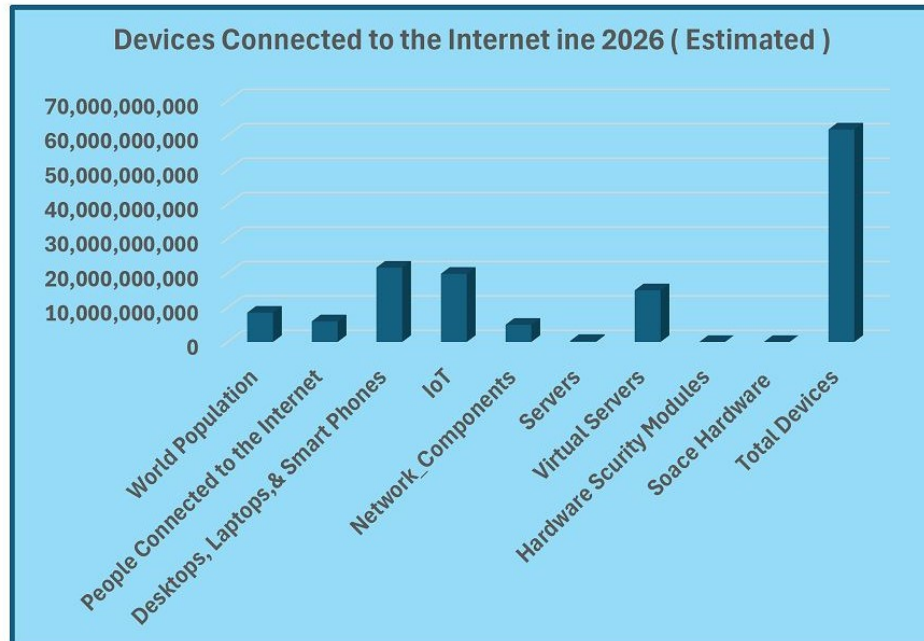


This Should Get Your Attention If You Care About PQC and the Internet

PQC System Engineer Readiness How Large Is This Opportunity?

World Population	8,500,000,000
People Connected to the Internet	6,000,000,000
Desktops, Laptops, & Smart Phones	21,600,000,000
IoT	19,800,000,000
Network Components	5,000,000,000
Servers	300,000,000
Virtual Servers	15,000,000,000
Hardware Security Modules	1,000,000
Soace Hardware	14,000
Total Devices	61,701,014,000

(All figures are estimated)



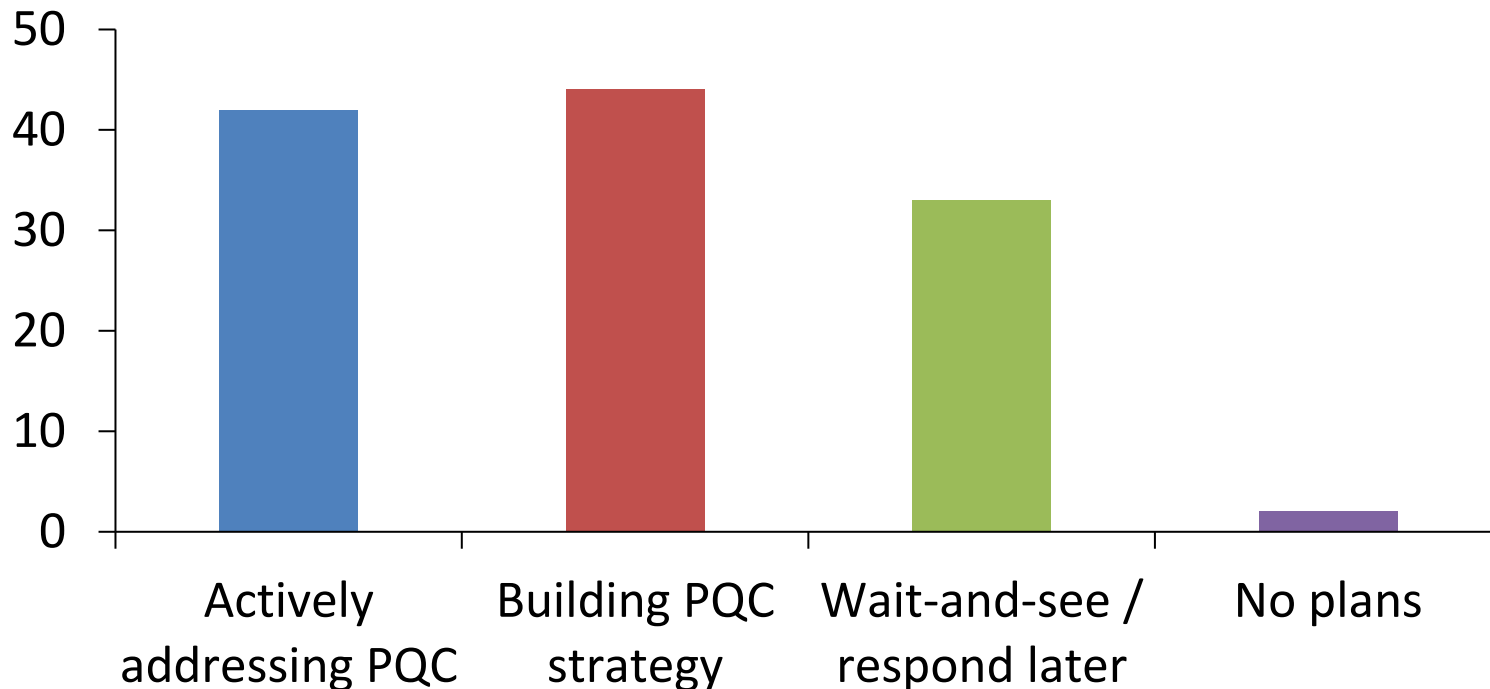
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Sunday, April 12, 2026
Chicago, Illinois, USA

This Should Get Your Attention If You Care About PQC and the Internet



Illustrative PQC Readiness

(Selected Survey Metrics)



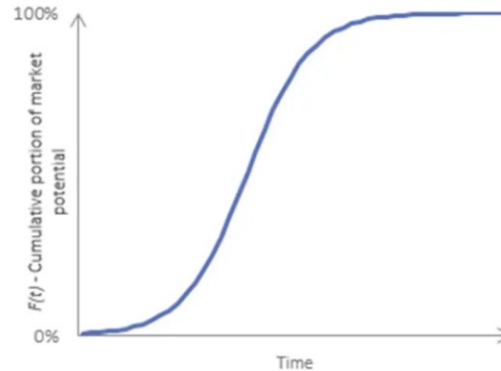
Note: Values are illustrative and based on multiple industry surveys.
Many large organizations are still in early planning stages.

Equation

$$\frac{f(t)}{1 - F(t)} = p + \frac{q}{m} A(t)$$

- m – market potential
- p – coefficient of innovation
- q – coefficient of imitation
- $f(t)$ – the portion of the market potential that adopts at time t
- $F(t)$ – the cumulative portion of the market potential that has adopted at time t

Representative Curve



Bass Diffusion Model

Who Has Started Their PQC Migrations?

🔒 The Quantum Deadline: Good, Bad & Ugly of PQC Adoption

By 2033, a cryptographically relevant quantum computer (CRQC) could emerge - yet most systems won't be ready.

Using the Bass model ($p \approx 0.03$, $q \approx 0.32$), we forecast:

~50% PQC adoption by 2033

Group 1 (defense, core infra): ~70% ready

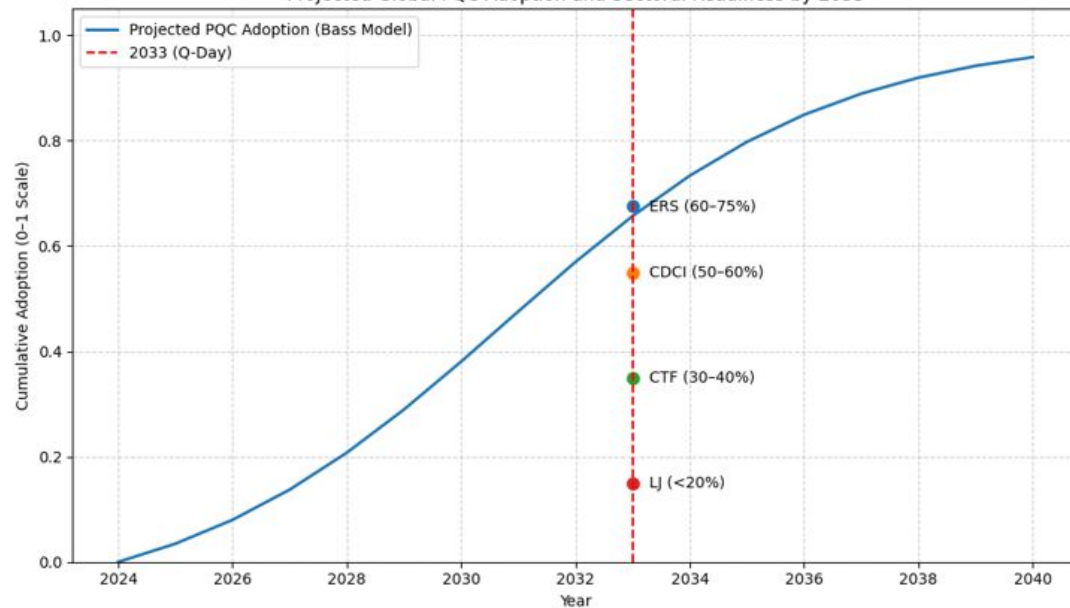
Group 2 (critical sectors): ~55%

Group 3 (commercial tech): ~35%

Group 4 (legacy systems): ~<20%

The risk? A fragmented, vulnerable internet—unless we act now.

Tehrani, M. (2025). "The Quantum Deadline: The Good, The Bad, and The Ugly of PQC Adoption", published d at Medium and retrieved from:
https://medium.com/@madjid.tehrani_17692/the-quantum-deadline-the-good-the-bad-and-the-ugly-of-pqc-adoption-baf0348812e4 .



PQC Readiness

 The Quantum Deadline: Good, Bad & Ugly of PQC Adoption

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IonQ Roadmap for Large-Scale, Fault-Tolerant Quantum Computers

Built on pioneering trapped ion research, IonQ's roadmap is enabled by the nearly 1,000 patented hardware and software breakthroughs that were developed at IonQ, integrated from strategic acquisitions.



Original IonQ #AQ Roadmap

2024	2025	2026	2027	2028
36 Algorithmic Qubits	64 Algorithmic Qubits	256 Algorithmic Qubits	384 Algorithmic Qubits	1,024 Algorithmic Qubits

Updated Technology Development Roadmap

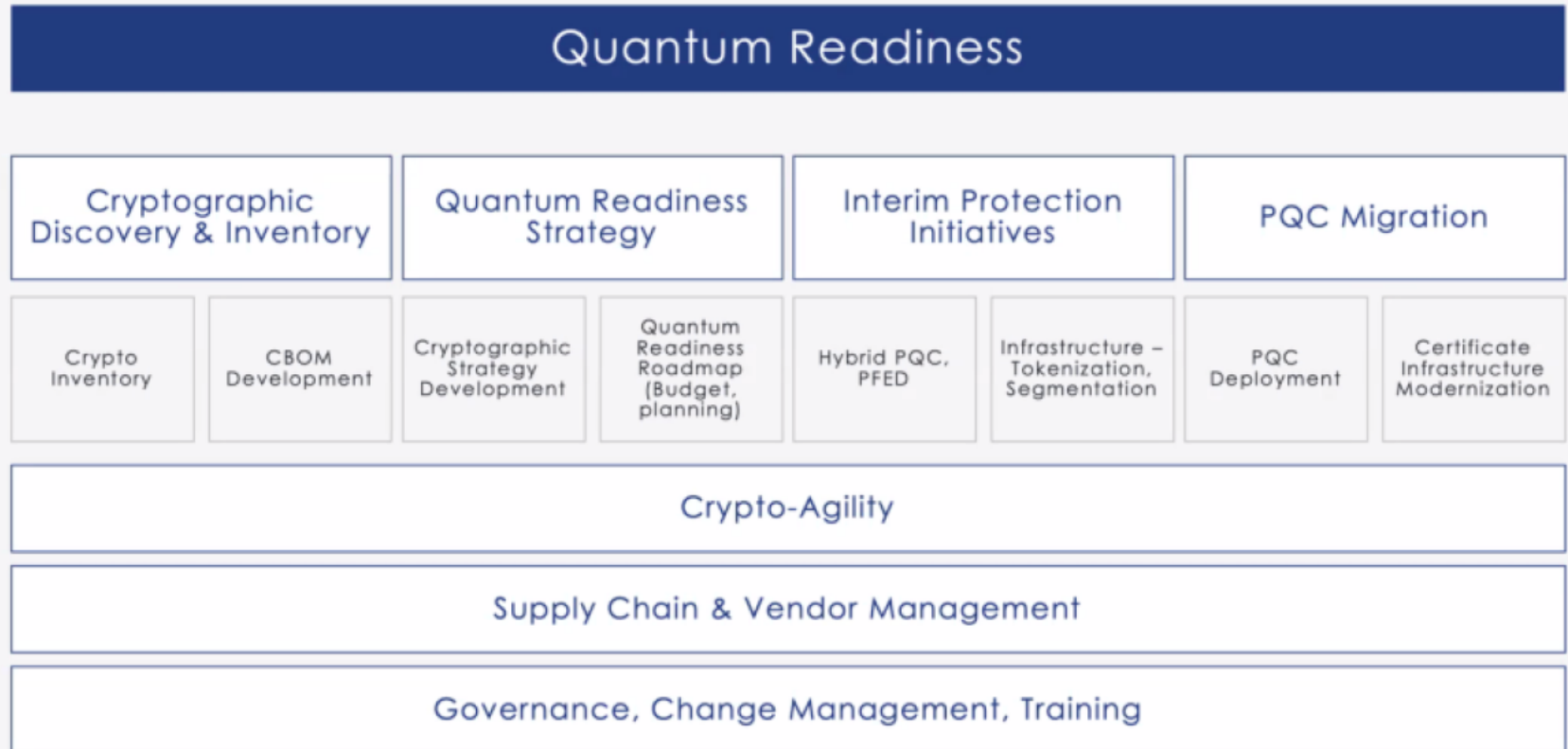
2024	2025	2026	2027	2028	2029	2030
36+ Physical Qubits 99.99% Physical Qubit Fidelity All-to-All Connectivity Optical Gate Operations 2D Qubit Array	64-100+ Physical Qubits 99.99% Physical Qubit Fidelity All-to-All Connectivity Microwave Gate Operations 2D Qubit Array Mid-Circuit Measurement Parallel Operations	100-256+ Physical Qubits 99.99% Physical Qubit Fidelity 12 Logical Qubits <1.00E-7 Logical Error Rate All-to-All Connectivity Microwave Gate Operations 2D Qubit Array Mid-Circuit Measurement Parallel Operations	10,000 Physical Qubits 99.99% Physical Qubit Fidelity 800 Logical Qubits <1.00E-7 Logical Error Rate All-to-All Connectivity Microwave Gate Operations 2D Qubit Array Mid-Circuit Measurement Parallel Operations	20,000 Physical Qubits 99.99% Physical Qubit Fidelity 1,600 Logical Qubits <1.00E-7 Logical Error Rate All-to-All Connectivity Microwave Gate Operations 2D Qubit Array Mid-Circuit Measurement Photonic Interconnect	200,000 Physical Qubits 99.99% Physical Qubit Fidelity 8,000 Logical Qubits <1.00E-12 Logical Error Rate All-to-All Connectivity Microwave Gate Operations 2D Qubit Array Mid-Circuit Measurement Photonic Interconnect	2,000,000 Physical Qubits 99.99% Physical Qubit Fidelity 80,000 Logical Qubits <1.00E-12 Logical Error Rate All-to-All Connectivity Microwave Gate Operations 2D Qubit Array Mid-Circuit Measurement Photonic Interconnect

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https://medium.com/@madjid.tehrani_17692/the-quantum-deadline-the-good-the-bad-and-the-ugly-of-pqc-adoption-baf0348812e4 .

Key Mitigation Steps to Reduce HNDL Risk

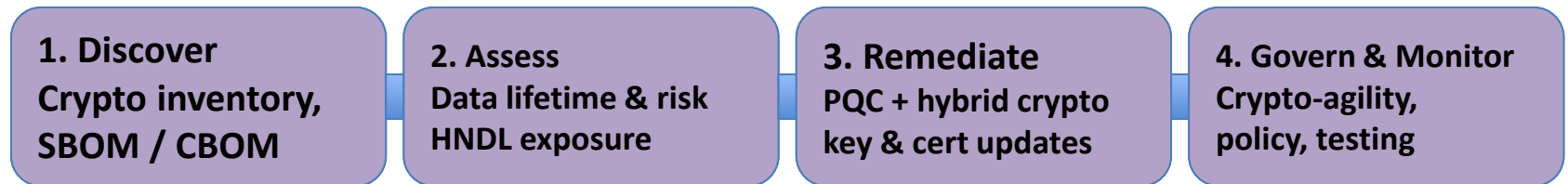
- Create a **cryptographic inventory** (**CBOM**) (where public-key crypto is used, which algorithms, key sizes, and data lifetime).
- Identify "long-lived" data and high-value targets (citizen data, health, finance, critical infrastructure, defense).
- Upgrade protocols (**TLS/VPN/email/IoT/PKI**) to **PQC** and, in the near term, hybrid (classical + PQC) modes.
- Implement **crypto-agility** so algorithms, key lengths, and certificates can be swapped quickly without redesign.
- Encrypt sensitive **data at rest** with quantum-safe keys and strong access controls.
- Continuously monitor standards, vendor roadmaps, and quantum threat intelligence.

Quantum Readiness



Slide Source: Mr. Marin Ivezic, 2025.

HNDL Mitigation Lifecycle



Cryptographic Agility is a measure of an organization's ability to adapt cryptographic solutions or algorithms (including their parameters and keys) quickly and efficiently in response to developments in cryptanalysis, emerging threats, technological advances, and/or vulnerabilities.

It is also a design principle for implementing, updating, replacing, running, and adapting cryptography and related business processes and policies with no significant architectural changes, minimal disruption to business operations, and short transition time.

- Design principle
- Adapt to ANY future threats to cryptography
- Minimal disruption to business

Good old exit plans, no more set and forget in cryptography

PKI Consortium (2025). PQC Conference, January 2025.

Cryptographic Agility

- ❖ All organizations need to be thinking about **Cryptographic Agility** RIGHT NOW. In fact, the experts are saying that all initial PQC Migration Projects MUST be coupled with Cryptographic Agility Projects that are conducted simultaneously.
- ❖ References:
 - ❖ **RFC 7696** <https://www.rfc-editor.org/rfc/pdf/rfc7696.txt.pdf>
 - ❖ ***Becoming Quantum Safe***, by Jai Singh Arun, Ray Harishankar, and Walid Rjaiibi, 2025

5-Year Crypto Agility Roadmap

<u>Year</u>	<u>Objective</u>
Year 1	Enterprise crypto inventory & CBOM
Year 2	Hybrid cryptography deployment
Year 3	Enterprise PQC migration scaling
Year 4	Legacy crypto retirement
Year 5	Automated crypto agility operations

Long-Term End State

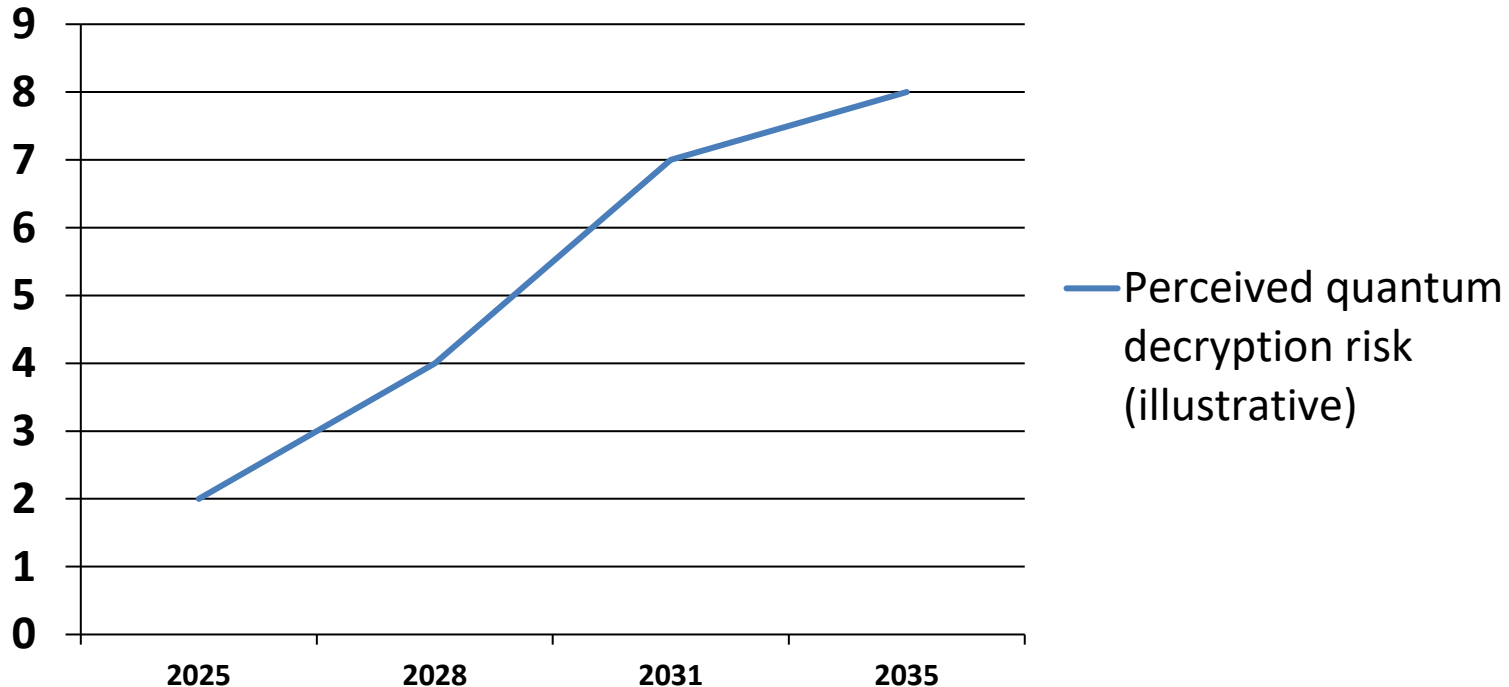
- Quantum-safe enterprise
- Fully automated certificate lifecycle
- Dynamic algorithm replacement
- Continuous crypto compliance
- Enterprise crypto telemetry & monitoring

Figure 1. PQC Roadmap Categories.



PKI Consortium (2025). PQC Conference, January 2025.

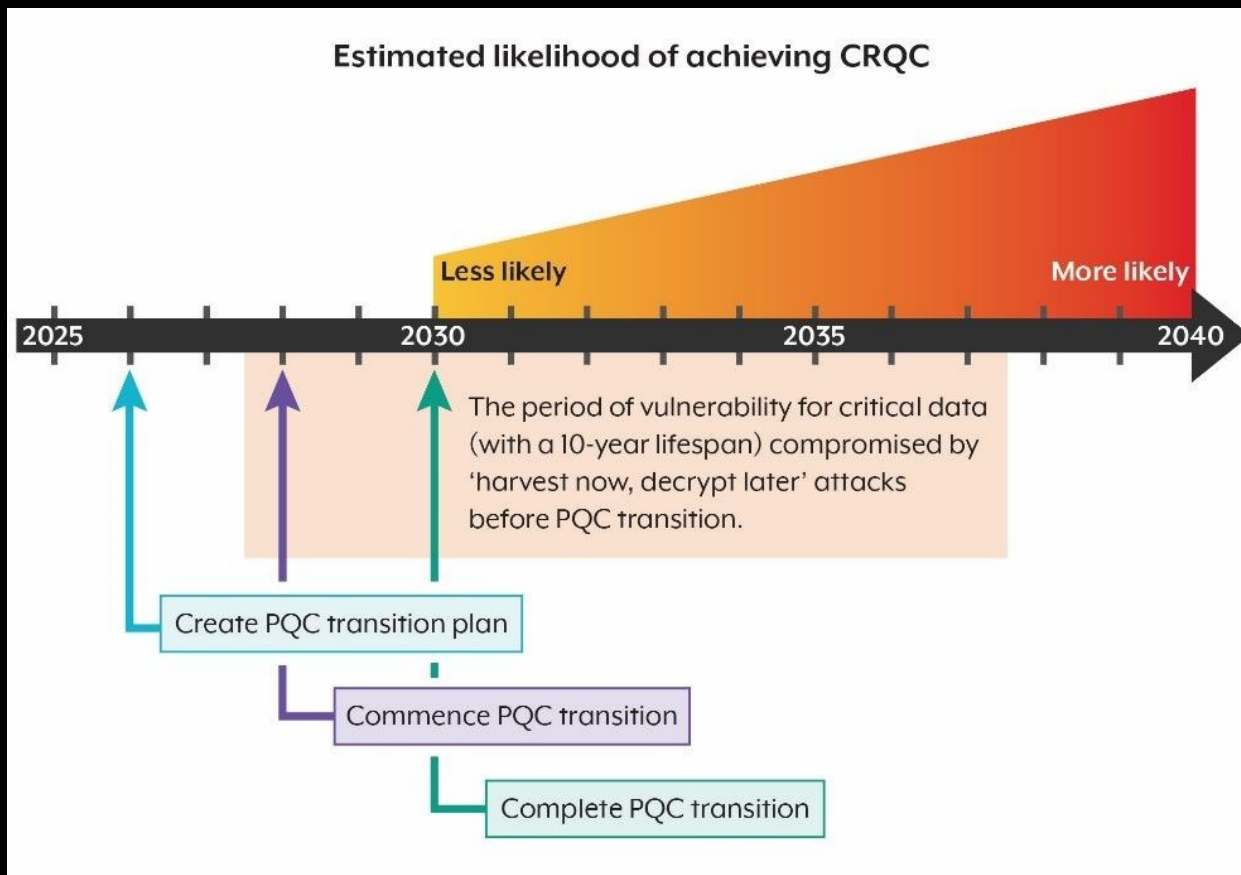
Illustrative Quantum Risk Timeline (2025–2035)



Analysts expect CRQCs capable of breaking today's public-key cryptography sometime in the 2030s, with some forecasts as early as ~2030–2035.

Because of HNDL, sensitive data must be protected well before "Q-Day".

Illustrative Quantum Risk Timeline (2025–2035)



Analysts expect CRQCs capable of breaking today's public-key cryptography sometime between 2029 and 2035, with some forecasts as early as ~2030–2035. Because of HNDL, sensitive data must be protected well before "Q-Day".

Key Takeaways & Call to Action

- **HNDL** is happening now: adversaries may already be collecting your encrypted data.
- The arrival of **CRQCs** is uncertain, but many experts place **Q-Day** between 2029 and 2033.
- Because of long data lifetimes, waiting until Q-Day is too late—historic data will already be exposed.
- Begin **PQC Planning** and **Migration** immediately, starting with discovery, prioritization, and crypto-agility.
- Treat **PQC Migration** as a strategic, multi-year program, not a one-time IT upgrade.
- Strive for **Cryptographic Agility** to become more effective at maintaining Cryptographic Requirements and Compliance in the Future.

HNDL

Questions and Answers



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HNDL

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HNDL

About Me

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3. CEO and owner of Slater Technologies, Inc. (since 1989)
4. In Quantum since 2011
5. Started in Cybersecurity in the U.S. Air Force: 1977.
6. Former U.S. Air Force Computer System Staff Officer who served at Strategic Air Command HQ.
7. Programming Languages: 43
8. 3 Graduate Degrees
 - MS Computer Information Systems
 - MBA
 - MS Cybersecurity
9. 88 Certifications including CISSP, SSCP, CISA, PMP.
10. Certified Project Manager with over 25 years experience.
11. Born and Raised in Memphis, Tennessee
12. Lives and works in Chicago, Illinois, USA
13. Published Author: <https://billslater.com/writing>
14. Working with QSECDEF since September 2024.
15. Happily married to Joanna Roguska since December 2000.
16. Avid fan of Andreas Vollenweider Music (since 1985), Chess (since 1964), and Judo (since 1968).
17. Voracious Reader with 18,000 physical books and 8000 digital books.
18. Meyers-Briggs: INFP.
19. Non-Denominational Christian (since 1973).



Sunrise, January 1, 2020
Uluru, Australia

HAPPY Q-DAY

Warning: It'll be the worst holiday in the history of the world.



BY AMIT KATWALA

ILLUSTRATIONS BY NICHOLAS LAW

Q Day is the finish line for PQC readiness



digitally remastered

**IT'S NOW
OR NEVER**

ELVIS PRESLEY

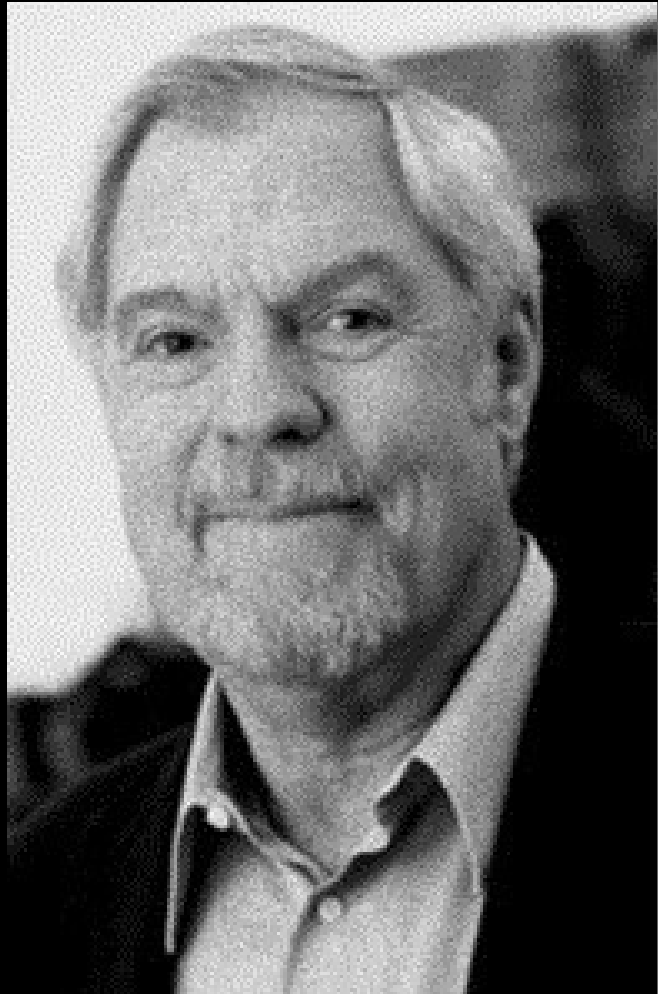
Supplemental Slides

Stockdale Paradox

You must never confuse faith that you will prevail in the end -- which you can never afford to lose -- with the discipline to confront the most brutal facts of your current reality, whatever they might be.

Admiral James B. Stockdale





"One of life's most painful moments comes when we must admit that we didn't do our homework, that we are not prepared."

Merlin Olsen

*I have learned
that people will
forget what you
said, people will
forget what you
did, but people
will never forget
how you made
them feel.*

Maya Angelou
1928-2014



HNDL

Photo by Michael Collopy

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Event Date and Time - Tuesday, April 7, 2026

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Steve Vaile is the Director. steve@qsecdef.com



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Muria Roberts is the Director - Muria.Roberts@protonmail.com



Chicago Quantum Meetup - Monthly Meetings

<https://www.meetup.com/chicago-quantum-computing-meetup-group-revival/>

Registration required - Meetup

Do an Internet Search - Chicago Quantum Meetup. Sometimes they have 1 to 0 meetings per month.

Barry Burd and Ron Schreiner are the Directors.



Washington DC Quantum Meetup - Monthly Meetings

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Helen Ma is the Director.



Alexandria, Egypt - Amsterdam, Netherlands Quantum Group - Monthly Meetings

<https://duckduckgo.com/?q=Alexandria%2C+Egypt+-+Amsterdam++Quantum+Group&atb=v314-1&ia=web>

Registration required on LinkedIn

Do an Internet Search - Alexandria Quantum Meetup. Sometimes they have 1 meetings per month, or every other Month.

Dr. Taha Selim is the Director.



Philippine Quantum Society

On Facebook, with announced Meetings on Zoom.

<https://www.facebook.com/qcsp.ph>

See the Quantum-Readiness Workshop, Apr 23-24:

<https://quantumreadyph.qcsp.institute>



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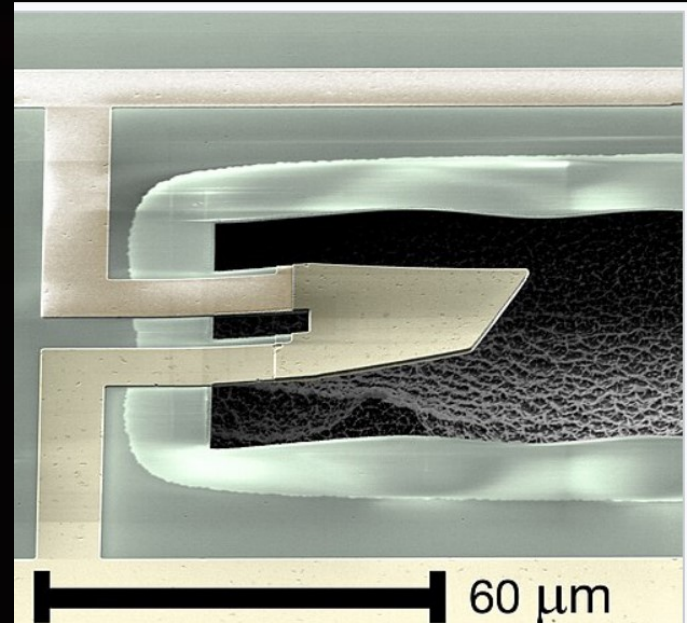
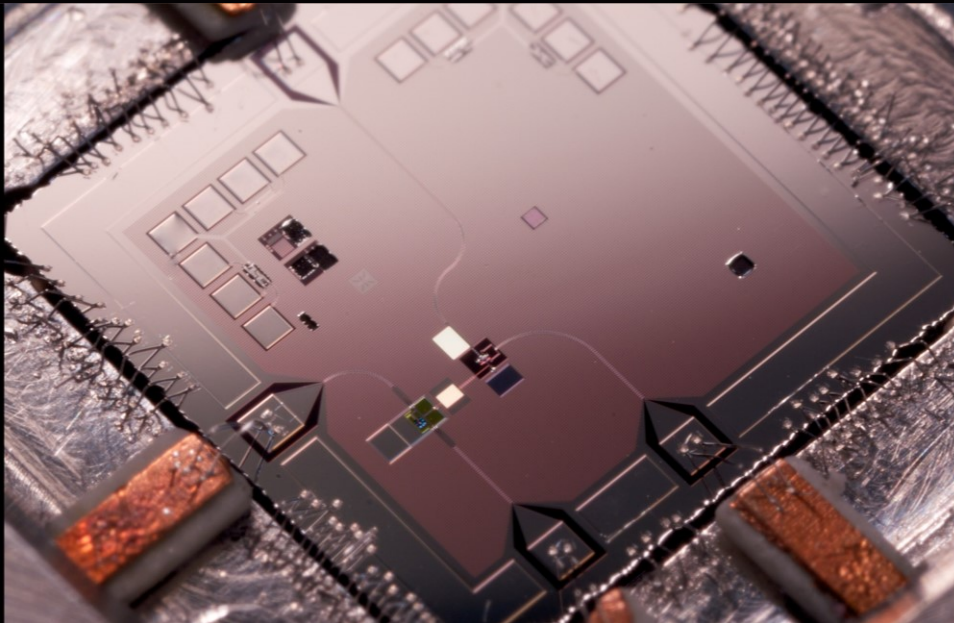
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PQC System Engineer Readiness

World's First Quantum Machine, 2010

Wikipedia, 2011 - https://en.wikipedia.org/wiki/Quantum_machine



End of Presentation

THE MAP OF QUANTUM PHYSICS

PRE-QUANTUM MYSTERIES

ATOMIC SPECTRA



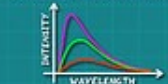
THE STABLE ATOM



RADIOACTIVITY



BLACKBODY RADIATION



PHOTOELECTRIC EFFECT



QUANTUM FOUNDATIONS

THE WAVE FUNCTION



PARTICLE-WAVE DUALITY



THE BORN RULE



INTERFERENCE PATTERN



THE SCHRÖDINGER EQUATION

$$i\hbar \frac{\partial}{\partial t} \Psi = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V(\mathbf{r}) \Psi$$

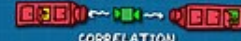
THE DIRAC EQUATION

$$i\hbar \gamma^\mu \partial_\mu \psi - mc\psi = 0$$

THE HEISENBERG UNCERTAINTY PRINCIPLE



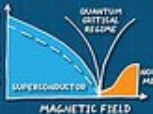
BELL'S THEOREM



ENERGY QUANTIZATION



PHASE TRANSITIONS



ZERO-POINT ENERGY



CASIMIR EFFECT



TOPOLOGICAL PHASE TRANSITIONS



MAGNETIC RESONANCE IMAGING



ATOMIC CLOCKS



ELECTRON MICROSCOPES



SCANNING TUNNELING MICROSCOPES

ATOMIC FORCE MICROSCOPES

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